

Summary and Recommendations

Customer Churn Analysis

Project Overview

This project analyzes customer churn behavior using a telecom customer dataset containing **7,043 customer records**. The objective was to identify the major factors influencing customer churn through data cleaning, exploratory data analysis (EDA), and business-focused visualizations. The analysis was performed using **Python, Pandas, NumPy, Matplotlib, and Seaborn**.

Key Findings

- Out of **7,043 customers**, **26.5%** discontinued the service, while **73.5%** remained with the company.
 - Customer **gender showed minimal impact** on churn, with similar churn behavior observed across both male and female customers.
 - Although **Senior Citizens represented only 16.2%** of the customer base, they experienced a noticeably higher churn rate than non-senior customers.
 - Customers with **Month-to-Month contracts** showed the highest churn, whereas **One-Year** and **Two-Year contracts** demonstrated significantly stronger customer retention.
 - Customers with **shorter tenure** were considerably more likely to churn, while long-term customers showed higher loyalty.
 - Customers **without Online Security and Tech Support services** were more likely to leave compared to those subscribed to these services.
 - **Electronic Check** users experienced the highest churn among all payment methods, while automatic payment methods showed better retention.
 - Additional services such as **Online Backup, Device Protection, Streaming TV, and Streaming Movies** also contributed positively to customer retention.
-

Business Recommendations

- Encourage customers to switch from **Month-to-Month** to long-term contracts through promotional offers.
- Focus retention campaigns on **new customers**, as churn is highest during the early stages of the customer lifecycle.

- Increase adoption of **Online Security** and **Tech Support** by offering bundled service packages.
 - Promote **automatic payment methods** to improve customer retention.
 - Develop targeted retention strategies for **high-risk customer segments**, including senior citizens and short-tenure customers.
-

Conclusion

The analysis demonstrates that **contract type, customer tenure, value-added services, and payment methods** have a stronger influence on customer churn than demographic characteristics. These insights enable businesses to identify high-risk customers, improve retention strategies, and reduce revenue loss through data-driven decision-making.